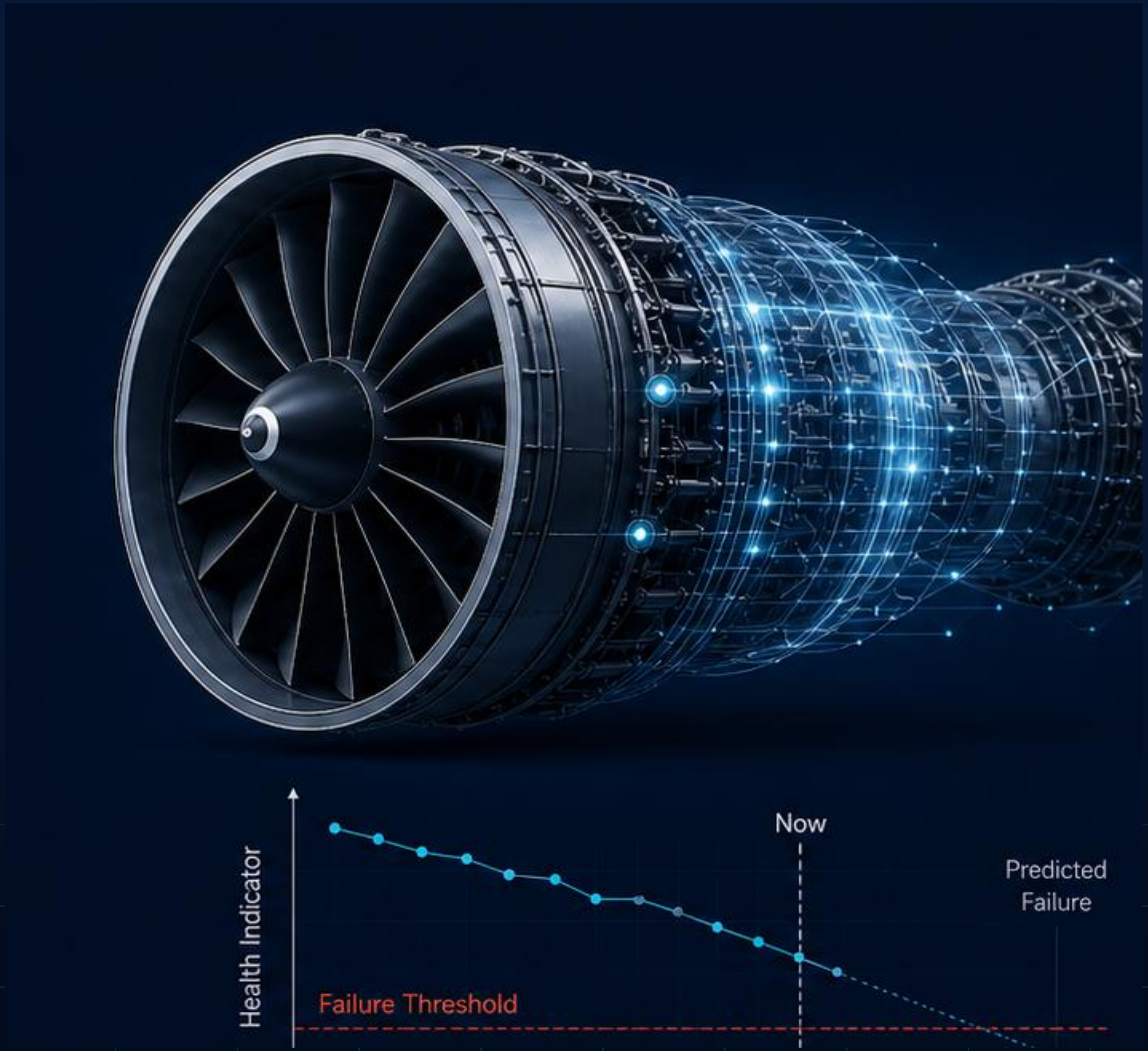
PROJECT VISION

Remaining Useful Life Prediction for Aerospace Technical Systems

- Predicting how many operating cycles remain before failure
- Working with multivariate sensor time series from technical systems
- Framing predictive maintenance as a time-series regression problem
- Connecting AI methods with reliability, safety and engineering decision-making

AI can support maintenance decisions before degradation becomes critical



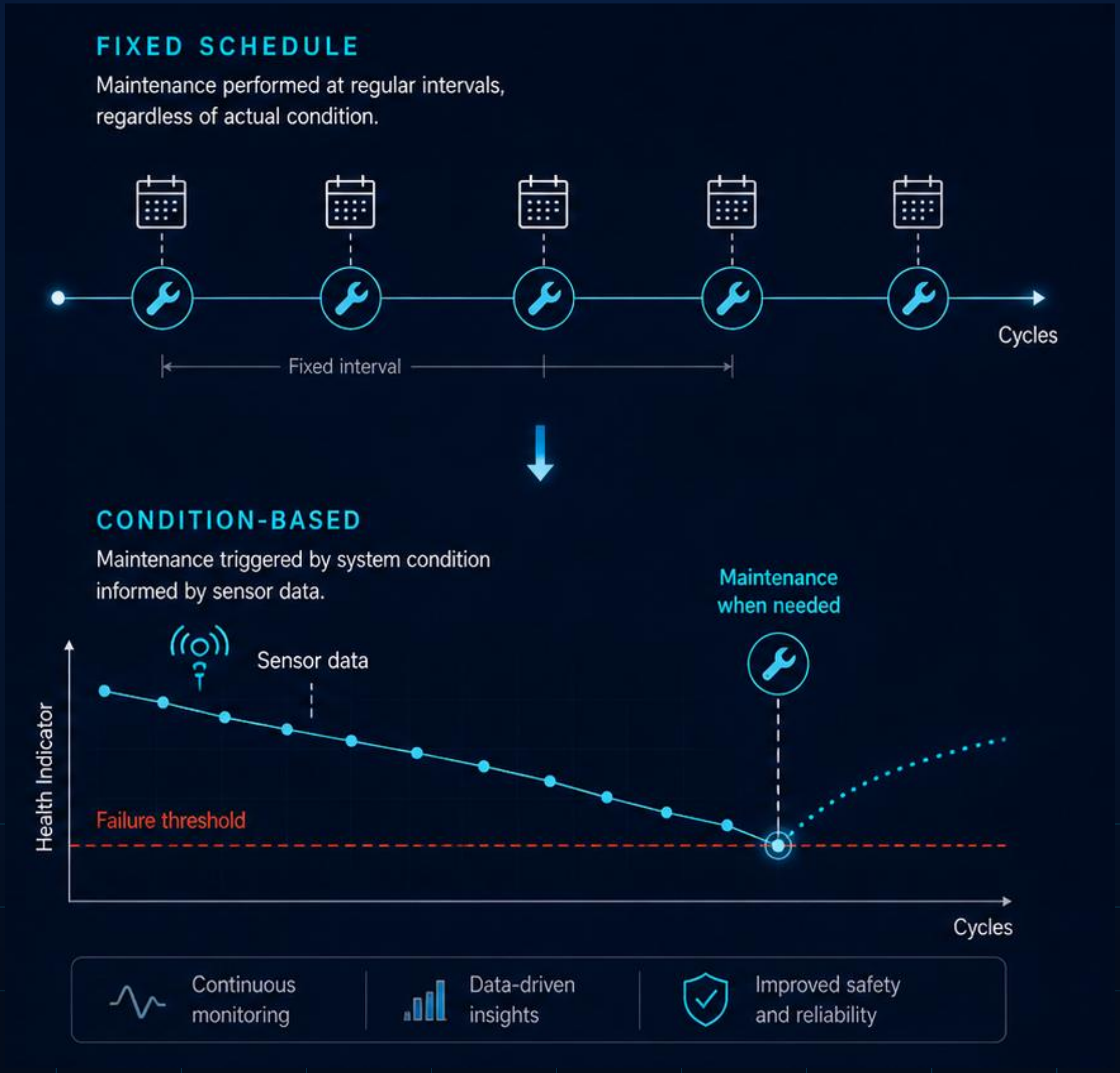
WHY IT MATTERS

Why Predictive Maintenance Matters

Maintenance Matters

- Aerospace systems operate under high safety and reliability requirements
- Scheduled maintenance can be costly because components are often replaced early to preserve large safety margins
- Real failure data is difficult to obtain because failures are rare, confidential and often prevented by scheduled service
- Remaining Useful Life (RUL) prediction uses sensor data to estimate when maintenance is actually needed

Predictive maintenance shifts decisions from fixed schedules to actual system condition.



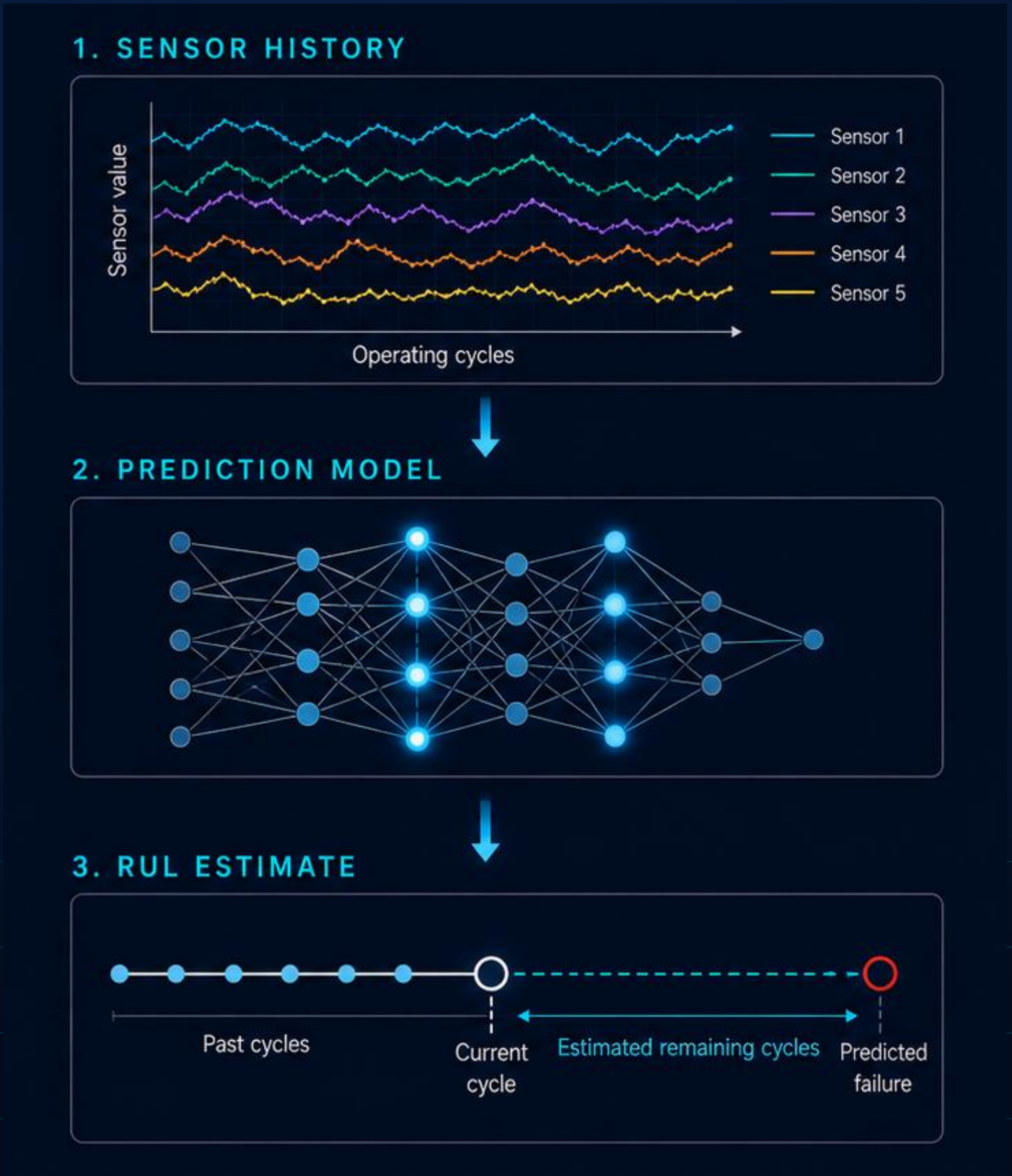
TASK FORMULATION

From Sensor History to RUL Prediction

to RUL Prediction

- Input: a time window of historical sensor measurements
- Data format: multivariate time series recorded across operating cycles
- Output: estimated Remaining Useful Life (RUL)
- Target: the true number of cycles remaining before failure
- Learning task: supervised regression

The model transforms multivariate sensor history into a numerical estimate of remaining system life



DATA STRATEGY

Benchmark Data for a Real Engineering Question

Real Engineering Question

- The project uses a public benchmark dataset of turbofan engine degradation
- Each engine is represented by operating conditions and sensor readings recorded over time
- Training trajectories contain complete run-to-failure histories
- This setup enables controlled comparison of different RUL prediction approaches

A public benchmark makes the engineering question measurable, reproducible and comparable

Dataset: NASA C-MAPSS FD001

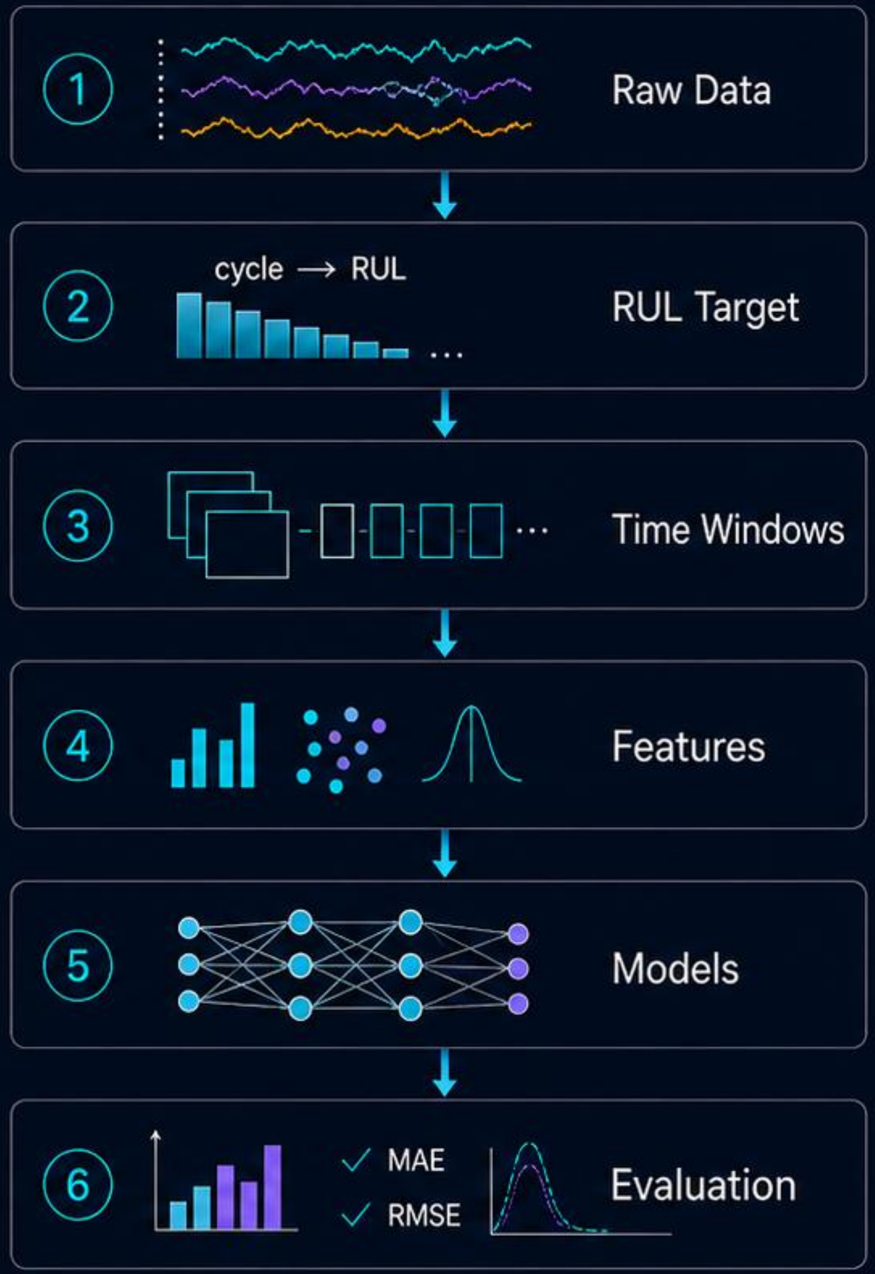
METHODOLOGICAL PIPELINE

A Reproducible RUL Prediction Pipeline

RUL Prediction Pipeline

- Load and structure multivariate time-series data
- Construct the RUL target for each engine cycle
- Build time-window representations
- Extract statistical and dynamic features
- Train baseline and nonlinear models
- Evaluate predictions with error metrics and degradation-stage analysis

The project is built as a full pipeline from raw sensor data to reproducible model evaluation



EXPECTED INSIGHTS

Expected Insights and Project Value

and Project Value

The project is expected to clarify:

- Which sensor patterns are most informative for RUL prediction?
- How much temporal history should a model use?
- When are simple models sufficient and when are nonlinear models necessary?
- At which degradation stages are predictions most reliable?
- How can model errors be interpreted for maintenance decisions?

The project goes beyond simple failure prediction by showing how degradation becomes visible in sensor data